

Intelligent Battery Management and Energy Saving System for Electric Vehicles

Problem Statement

Electric vehicle riders may sometimes travel long distances without realizing that the battery level is becoming critically low.

When the battery charge falls below a certain level:

- High-speed driving consumes more power.
- Rapid acceleration drains the battery quickly.
- Non-essential loads waste energy.
- The rider may be stranded before reaching the destination.

Therefore, an intelligent system is required to automatically reduce power consumption and extend the remaining driving range.

Project Objective

The objectives of the project are:

- Continuously monitor battery percentage.
- Detect low battery conditions.
- Automatically activate Energy Saving Mode.
- Limit vehicle speed.
- Reduce unnecessary power consumption.
- Increase the remaining travel range.
- Warn the rider about low battery status.

Working Principle

The system continuously monitors:

- Battery percentage
- Vehicle speed
- Power consumption

- Auxiliary electrical loads

When the battery level falls below a predefined threshold, the ESP32 automatically activates Energy Saving Mode.

The controller then:

- Limits maximum vehicle speed.
- Restricts acceleration.
- Reduces motor power output.
- Turns off unnecessary electrical loads.
- Displays battery-saving warnings.

This helps the rider travel a longer distance with the remaining battery.

Components Required

Component	Quantity
ESP32 Controller	1
Battery Voltage Sensor	1
Current Sensor (ACS712/INA219)	1
Motor Controller (ESC)	1
OLED/TFT Display	1
Buzzer	1
Throttle Sensor	1
Battery Pack	1

System Parameters Monitored

The system continuously measures:

Battery Percentage

Battery = 100% to 0%

Battery Voltage

48V Battery System

Current Consumption

Motor Current

Vehicle Speed

km/h

Battery Level Classification

Battery Level Vehicle Mode	
100% - 50%	Normal Mode
50% - 30%	Eco Mode
Below 30%	Power Saving Mode
Below 15%	Critical Mode
Below 5%	Emergency Mode

System Operation

Normal Mode

Battery:

100% - 50%

Vehicle operates normally.

- Full speed available.
- Full acceleration available.
- All features active.

Eco Mode

Battery:

50% - 30%

Controller automatically:

- Reduces acceleration.
- Limits peak motor current.
- Displays Eco Mode notification.

Display:

ECO MODE ACTIVE

Battery Saving Enabled

Power Saving Mode

Battery:

Below 30%

Controller activates battery protection.

Actions:

- Speed limited.
- Motor power reduced.
- Auxiliary loads disabled.
- Rider warned.

Display:

LOW BATTERY

POWER SAVING MODE

ACTIVE

Critical Battery Mode

Battery:

Below 15%

Additional restrictions applied.

- Strong acceleration disabled.
- Maximum power greatly reduced.
- Continuous warning displayed.

Display:

CRITICAL BATTERY

REACH CHARGING
STATION

Speed Limiting Logic

Example:

Normal Battery

Battery > 50%

Maximum Speed = 60 km/h

Eco Mode

Battery 30% - 50%

Maximum Speed = 40 km/h

Power Saving Mode

Battery < 30%

Maximum Speed = 20 km/h

Critical Mode

Battery < 15%

Maximum Speed = 10 km/h

Control Algorithm

IF Battery > 50%

Normal Mode

ELSE IF Battery <= 50%

AND Battery > 30%

Eco Mode

ELSE IF Battery $\leq$ 30%
AND Battery $>$ 15%

Power Saving Mode

ELSE IF Battery $\leq$ 15%

Critical Mode

Power Saving Features

Motor Power Limitation

The ESP32 sends commands to the motor controller to reduce:

- Peak current
- Torque
- Acceleration

This reduces battery consumption.

Throttle Limitation

Even if the rider fully twists the throttle:

100% Throttle Input

↓

Controller Allows Only

40% Motor Power

This prevents excessive energy usage.

Auxiliary Load Control

When battery becomes low:

Disable:

- Decorative lighting
- Extra displays
- USB charging ports
- Non-essential electronics

Only essential systems remain active.

Estimated Range Prediction

The system can estimate remaining distance.

Example:

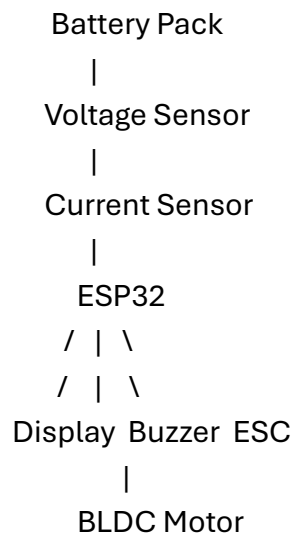
Battery : 25%

Estimated Range :

12 km Remaining

This helps the rider decide whether they can safely return home.

System Architecture



Flowchart

Start

|

Read Battery Level

|

Battery > 50% ?

|

Yes

|

Normal Mode

|

No

|

Battery > 30% ?

|

Yes

|

Eco Mode

|

No

|

Battery > 15% ?

|

Yes

|

Power Saving Mode

|

No

|

Critical Mode

|

Limit Speed

Reduce Power

Display Warning

|

Continue Monitoring

OLED Display Outputs

Normal Mode

Battery : 80%

Mode : Normal

Speed : 45 km/h

Eco Mode

Battery : 40%

Mode : ECO

Speed Limited

Power Saving Mode

Battery : 25%

POWER SAVING

Max Speed:

20 km/h

Critical Mode

Battery : 10%

CRITICAL BATTERY

Max Speed:

10 km/h

Charge Soon

Conclusion

The **Intelligent Battery Management and Energy Saving System** continuously monitors battery status and automatically activates different power-saving levels based on the remaining charge. By limiting speed, reducing motor power, restricting acceleration, and disabling unnecessary loads, the system significantly extends the remaining driving range and helps the rider safely reach a charging station or home before the battery is fully depleted. This project combines battery management, embedded control, power electronics, and automotive automation, making it an excellent EV-focused engineering project.